2.31 Improper Integrals

Worked Solutions

1 a)

The upper limit is ∞, so the integral has an infinite limit and is improper.

1 b)

The integrand $1/(3x^3)$ is undefined at $x = 0$, which is one of the limits, so the integral is improper.

2 a)

$$\int_2^\infty \frac{1}{x^2}\,dx = \lim_{a\to\infty}\int_2^a x^{-2}\,dx = \lim_{a\to\infty}\left[-\frac{1}{x}\right]_2^a = \lim_{a\to\infty}\left(-\frac{1}{a}+\frac{1}{2}\right) = \frac{1}{2}.$$

2 b)

$$\int_2^\infty \frac{1}{x^3}\,dx = \lim_{a\to\infty}\left[-\frac{1}{2x^2}\right]_2^a = \lim_{a\to\infty}\left(-\frac{1}{2a^2}+\frac{1}{8}\right) = \frac{1}{8}.$$

2 c)

$$\int_2^\infty \frac{1}{x^4}\,dx = \lim_{a\to\infty}\left[-\frac{1}{3x^3}\right]_2^a = \lim_{a\to\infty}\left(-\frac{1}{3a^3}+\frac{1}{24}\right) = \frac{1}{24}.$$

As $a \to \infty$, $1/a^2 \to 0$.

3 a)

$$\int_0^2 \frac{1}{x^2}\,dx = \lim_{a\to 0^+}\left[-\frac{1}{x}\right]_a^2 = \lim_{a\to 0^+}\left(-\frac{1}{2}+\frac{1}{a}\right) = +\infty,$$

so the integral diverges.

3 b)

$$\int_0^2 \frac{1}{x^3}\,dx = \lim_{a\to 0^+}\left[-\frac{1}{2x^2}\right]_a^2 = \lim_{a\to 0^+}\left(-\frac{1}{8}+\frac{1}{2a^2}\right) = +\infty.$$

Therefore the integral diverges.

3 c)

$$\int_0^2 \frac{1}{x^4}\,dx = \lim_{a\to 0^+}\left[-\frac{1}{3x^3}\right]_a^2 = \lim_{a\to 0^+}\left(-\frac{1}{24}+\frac{1}{3a^3}\right) = +\infty,$$

so the integral diverges.

4 a)

$$\int_0^9 \frac{1}{\sqrt{x}}\,dx = \lim_{a\to 0^+}\left[2\sqrt{x}\right]_a^9 = \lim_{a\to 0^+}(6-2\sqrt{a}) = 6.$$

4 b)

$$\int_4^\infty \frac{1}{\sqrt{x}}\,dx = \lim_{a\to\infty}\left[2\sqrt{x}\right]_4^a = \lim_{a\to\infty}(2\sqrt{a}-4) = +\infty,$$

so the integral diverges.

5 a)

$$\int_0^8 x^{-3/4}\,dx = \lim_{a\to 0^+}\left[4x^{1/4}\right]_a^8 = 4\sqrt[4]{8}.$$

5 b)

$$\int_8^\infty x^{-3/4}\,dx = \lim_{a\to\infty}\left[4x^{1/4}\right]_8^a = +\infty,$$

so the integral diverges.

6 a)

$$\int_4^\infty 192x^{-4}\,dx = \lim_{a\to\infty}\left[-64x^{-3}\right]_4^a = 0-(-1) = 1.$$

6 b)

$$\int_4^\infty (192x^{-4}+1)\,dx = \lim_{a\to\infty}\left[-64x^{-3}+x\right]_4^a = +\infty,$$

so it diverges.

6 c)

$$\int_4^\infty 192x^{-4}(x+1)\,dx = \lim_{a\to\infty}\int_4^a (192x^{-3}+192x^{-4})\,dx = \lim_{a\to\infty}\left[-96x^{-2}-64x^{-3}\right]_4^a = 6+1 = 7.$$

7)

$$\int_0^4 \frac{1}{x^3\sqrt{x}}\,dx = \lim_{a\to 0^+}\int_a^4 x^{-7/2}\,dx = \lim_{a\to 0^+}\left[-\frac{2}{5}x^{-5/2}\right]_a^4 = +\infty,$$

so it diverges.

8)

$$\int_0^{1/64}\left(x^{-1/2}-x^{-1/3}\right)\,dx = \left[2\sqrt{x}-\frac{3}{2}x^{2/3}\right]_0^{1/64} = \frac{1}{4}-\frac{3}{32} = \frac{5}{32}.$$

9 a)

Amelia's method is incorrect because $35/x^6$ is undefined at $x=0$, so this is an improper integral. She should split it at a value a tending to zero, evaluate the two resulting integrals as limits, and then test whether both limits converge.

9 b)

$$\begin{aligned}\int_{-2}^{2} \frac{35}{x^6}\,dx &= \lim_{a\to 0^-}\int_{-2}^{a} 35x^{-6}\,dx + \lim_{a\to 0^+}\int_{a}^{2} 35x^{-6}\,dx \\ &= \lim_{a\to 0^-}\left[-7x^{-5}\right]_{-2}^{a} + \lim_{a\to 0^+}\left[-7x^{-5}\right]_{a}^{2} \\ &= \lim_{a\to 0^-}\left(-7a^{-5} - \frac{7}{32}\right) + \lim_{a\to 0^+}\left(-\frac{7}{32} + 7a^{-5}\right).\end{aligned}$$

As $a \to 0^-$, $-7a^{-5} \to +\infty$, and as $a \to 0^+$, $7a^{-5} \to +\infty$. The integral diverges.

10)

The integrand is undefined at $x = 0$, so split the integral there:

$$\begin{aligned}\int_{-1}^{8} \frac{4}{x^{2/3}}\,dx &= \lim_{a\to 0^-}\left[12x^{1/3}\right]_{-1}^{a} + \lim_{a\to 0^+}\left[12x^{1/3}\right]_{a}^{8} \\ &= \lim_{a\to 0^-}(12a^{1/3} + 12) + \lim_{a\to 0^+}(24 - 12a^{1/3}) \\ &= 12 + 24 = 36.\end{aligned}$$

11)

The integrand is undefined at $x = 0$, so split the integral there:

$$\begin{aligned}\int_{-32}^{1} \frac{100}{x^{4/5}}\,dx &= \lim_{a\to 0^-}\left[500x^{1/5}\right]_{-32}^{a} + \lim_{a\to 0^+}\left[500x^{1/5}\right]_{a}^{1} \\ &= \lim_{a\to 0^-}(500a^{1/5} + 1000) + \lim_{a\to 0^+}(500 - 500a^{1/5}) \\ &= 1000 + 500 = 1500.\end{aligned}$$

12 a)

$1/(x-5)$ is undefined at $x = 5$, which lies between the limits 4 and 6; hence this is improper.

12 b)

$e^x/(e^x - 2)$ is undefined when $e^x = 2$, or $x = \ln 2 \approx 0.693$, which is between 0 and 1; hence this is improper.

12 c)

The denominator vanishes when $\sin x - \sqrt{3}\cos x = 0$, so $\tan x = \sqrt{3}$ and $x = \pi/3$. This lies between 0 and $2\pi/3$, so the integral is improper.

12 d)

$\ln x$ is undefined at $x = 0$, which is one of the limits; hence the integral is improper.

13 a)

$$\int_{2}^{\infty} \frac{1}{x}\,dx = \lim_{a\to\infty}[\ln x]_2^a = \lim_{a\to\infty}(\ln a - \ln 2) = +\infty.$$

Therefore it diverges.

13 b)

$$\int_0^2 \frac{1}{x}\,dx = \lim_{a\to 0^+} [\ln x]_a^2 = \lim_{a\to 0^+} (\ln 2 - \ln a) = +\infty,$$

so it diverges.

14 a)

$$\int_0^\infty \frac{1}{e^x}\,dx = \lim_{a\to\infty} \left[-e^{-x}\right]_0^a = \lim_{a\to\infty} (1 - e^{-a}) = 1.$$

14 b)

$$\int_0^\infty \frac{1}{e^{2x}}\,dx = \lim_{a\to\infty} \left[-\frac{1}{2}e^{-2x}\right]_0^a = \frac{1}{2}.$$

15)

$$\int_7^\infty (x-3)^{-5/2}\,dx = \lim_{a\to\infty} \left[-\frac{2}{3}(x-3)^{-3/2}\right]_7^a = \frac{2}{3\cdot 8} = \frac{1}{12}.$$

16)

The integrand is undefined at $x = -3/2$, so split the interval there. The two limits contain the term

$$\int \frac{2}{(2x+3)^6}\,dx = -\frac{1}{5(2x+3)^5} + c.$$

As $x \to -3/2$ from either side, the corresponding improper integral tends to $+\infty$. Hence it diverges.

17 a)

$$\int_3^4 \frac{x}{x^2-9}\,dx = \lim_{a\to 3^+} \left[\frac{1}{2}\ln|x^2-9|\right]_a^4 = \lim_{a\to 3^+} \frac{1}{2}\Big(\ln 7 - \ln(a^2-9)\Big) = +\infty.$$

Thus it diverges.

17 b)

$$\int_3^4 \frac{x}{\sqrt{x^2-9}}\,dx = \lim_{a\to 3^+} \left[\sqrt{x^2-9}\right]_a^4 = \sqrt{7}.$$

18 a)

$$\int_0^\infty 4xe^{-x^2}\,dx = \lim_{a\to\infty} \left[-2e^{-x^2}\right]_0^a = 2.$$

18 b)

Split at zero:

$$\int_{-\infty}^\infty 4xe^{-x^2}\,dx = \lim_{a\to-\infty} \left[-2e^{-x^2}\right]_a^0 + \lim_{b\to\infty} \left[-2e^{-x^2}\right]_0^b = -2 + 2 = 0.$$

19)

$$\int_{-\infty}^{0} \frac{e^x}{e^x+1}\,dx = \lim_{a\to-\infty}\left[\ln(e^x+1)\right]_a^0 = \ln 2 - \lim_{a\to-\infty}\ln(1+e^a) = \ln 2.$$

20)

$$\int_0^{\pi/2} \tan x\,dx = \lim_{a\to(\pi/2)^-}\left[-\ln|\cos x|\right]_0^a = \lim_{a\to(\pi/2)^-} -\ln(\cos a) = +\infty.$$

It diverges.

21)

Since $8-2x-x^2 = 9-(x+1)^2$, the denominator is zero at the upper limit $x=2$. Using $u = 8-2x-x^2$,

$$\int_{-1}^{2} \frac{x+1}{\sqrt{8-2x-x^2}}\,dx = \lim_{a\to 2^-}\left[-\sqrt{8-2x-x^2}\right]_{-1}^{a} = 3.$$

22 a)

$$\lim_{a\to\infty}\frac{6a-5}{3a+7} = \lim_{a\to\infty}\frac{6-5/a}{3+7/a} = 2.$$

22 b)

$$\int_4^{\infty}\left(\frac{6}{6x-5}-\frac{3}{3x+7}\right)dx = \lim_{a\to\infty}\left[\ln|6x-5|-\ln|3x+7|\right]_4^a.$$

At $x=4$, both logarithm arguments equal 19; as $a\to\infty$, $(6a-5)/(3a+7)\to 2$ by part a. Thus the limit is $\ln 2$.

23)

Using partial fractions,

$$\frac{2}{x(3x+1)} = \frac{2}{x} - \frac{6}{3x+1}.$$

Hence

$$\int_1^{\infty}\frac{3x-2}{x(3x+1)}\,dx = \int_1^{\infty}\left(\frac{2}{x}-\frac{6}{3x+1}\right)dx = \lim_{a\to\infty}\left[2\ln x - 2\ln(3x+1)\right]_1^a = \ln\frac{16}{9},$$

so $k = 16/9$.

24)

Put $u=\sqrt{x+1}$, so $x = u^2-1$, $dx = 2u\,du$, and the limits become 0 and $\sqrt{3}$. Then

$$\int_{-1}^{2}\frac{2-x}{\sqrt{x+1}}\,dx = \int_0^{\sqrt{3}} 2(3-u^2)\,du = \left[6u-\frac{2}{3}u^3\right]_0^{\sqrt{3}} = 4\sqrt{3}.$$

25)

Let $u = 1/x$, so $dx = -du/u^2$ and the limits become π and 0:

$$\int_{1/\pi}^{\infty} \frac{\sin(1/x)}{x^2}\,dx = \int_0^{\pi} \sin u\,du = [-\cos u]_0^{\pi} = 2.$$

26)

Let $u = 2 - x^3e^{-2x}$. Then $du = -(3-2x)x^2e^{-2x}\,dx$ and

$$\frac{(3-2x)x^2}{x^3 - 2e^{2x}}\,dx = \frac{du}{u}.$$

The lower limit $x = 3/2$ gives $u = 2 - \frac{27}{8e^3} = \frac{16e^3-27}{8e^3}$, while $u \to 2$ as $x \to \infty$. Therefore

$$\int_{3/2}^{\infty} \frac{(3-2x)x^2}{x^3 - 2e^{2x}}\,dx = \ln 2 - \ln\left(\frac{16e^3-27}{8e^3}\right) = \ln\left(\frac{16e^3}{16e^3-27}\right).$$

27)

The integrand is undefined at $x = \pm 1$, so split at zero and approach both endpoints by limits:

$$\int_{-1}^{1} \frac{1}{\sqrt{1-x^2}}\,dx = \lim_{a\to -1^+}\int_a^0 \frac{dx}{\sqrt{1-x^2}} + \lim_{b\to 1^-}\int_0^b \frac{dx}{\sqrt{1-x^2}} = \frac{\pi}{2} + \frac{\pi}{2} = \pi.$$

28)

$$\int_0^{\infty} (\cosh x - \sinh x)\,dx = \lim_{a\to\infty}[\sinh x - \cosh x]_0^a = \lim_{a\to\infty}(1 - e^{-a}) = 1.$$

29 a)

Using partial fractions,

$$\frac{1}{4x^2-9} = \frac{1}{6(2x-3)} - \frac{1}{6(2x+3)}.$$

Thus

$$\int_2^{\infty} \frac{1}{4x^2-9}\,dx = \lim_{a\to\infty}\left[\frac{1}{12}\ln|2x-3| - \frac{1}{12}\ln|2x+3|\right]_2^a = \frac{1}{12}\ln 7.$$

29 b)

$$\int_0^{\infty} \frac{1}{4x^2+9}\,dx = \lim_{a\to\infty}\left[\frac{1}{6}\arctan\left(\frac{2x}{3}\right)\right]_0^a = \frac{\pi}{12}.$$

30)

Partial fractions give

$$\frac{12-6x}{(2x+3)(x^2+3)} = \frac{4}{2x+3} - \frac{2x}{x^2+3}.$$

Hence

$$\int_0^{\infty} \frac{12-6x}{(2x+3)(x^2+3)}\,dx = \lim_{a\to\infty}\Big[2\ln(2x+3) - \ln(x^2+3)\Big]_0^a = \ln 4,$$

so $k = 4$.

31)

Partial fractions give

$$\frac{3x^2+16}{x^2(x^2+4)} = \frac{4}{x^2} - \frac{1}{x^2+4}.$$

Therefore

$$\int_2^\infty \frac{3x^2+16}{x^2(x^2+4)}\,dx = \lim_{a\to\infty}\left[-\frac{4}{x} - \frac{1}{2}\arctan\left(\frac{x}{2}\right)\right]_2^a = 2 - \frac{\pi}{8},$$

so $k = 1/8$.

32)

The shaded region is between $x = 1$ and $x = 2$. The improper integral at $x = 1$ is convergent:

$$A = \lim_{a\to 1^+}\int_a^2 \frac{1}{\sqrt{x^2-1}}\,dx = \lim_{a\to 1^+}\left[\operatorname{arcosh} x\right]_a^2 = \ln(2+\sqrt{3}).$$

33)

Complete the square: $x^2 + 8x + 17 = (x+4)^2 + 1$. The total shaded area is

$$\int_{-\infty}^\infty \frac{dx}{(x+4)^2+1} = \lim_{a\to-\infty}\left[\arctan(x+4)\right]_a^0 + \lim_{b\to\infty}\left[\arctan(x+4)\right]_0^b = \pi.$$

34)

$$\int_0^1 \frac{\ln x}{\sqrt{x}}\,dx = \lim_{a\to 0^+}\int_a^1 x^{-1/2}\ln x\,dx.$$

By parts with $u = \ln x$, $dv = x^{-1/2}dx$, an antiderivative is $2\sqrt{x}\ln x - 4\sqrt{x}$. Hence the limit is -4 (since $\sqrt{a}\ln a \to 0$).

35)

$$\int_0^{12} \ln x\,dx = \lim_{a\to 0^+}\left[x\ln x - x\right]_a^{12} = 12\ln 12 - 12,$$

using $\lim_{a\to 0^+} a\ln a = 0$.

36)

Integrate by parts twice:

$$\int x^2 e^{-3x}\,dx = -\frac{1}{3}x^2e^{-3x} - \frac{2}{9}xe^{-3x} - \frac{2}{27}e^{-3x} + c.$$

Thus

$$\int_0^\infty x^2e^{-3x}\,dx = \lim_{a\to\infty}\left[-\frac{1}{3}x^2e^{-3x} - \frac{2}{9}xe^{-3x} - \frac{2}{27}e^{-3x}\right]_0^a = \frac{2}{27}.$$

37 a)

By L'Hôpital's Rule,

$$\lim_{x\to\infty}\frac{\ln x}{x} = \lim_{x\to\infty}\frac{1/x}{1} = 0.$$

37 b)

By parts, an antiderivative is $-\ln x/x - 1/x$. Therefore

$$\int_1^\infty \frac{\ln x}{x^2}\,dx = \lim_{a\to\infty}\left[-\frac{\ln x}{x} - \frac{1}{x}\right]_1^a = 1,$$

using part a).

38 a)

By L'Hôpital's Rule,

$$\lim_{x\to\infty}\frac{x}{e^x} = \lim_{x\to\infty}\frac{1}{e^x} = 0.$$

38 b)

$$\int_{-\infty}^0 xe^x\,dx = \lim_{a\to-\infty}[xe^x - e^x]_a^0 = -1 - \lim_{a\to-\infty}(ae^a - e^a) = -1,$$

since $e^a \to 0$ and $ae^a = a/e^{-a} \to 0$ by part a).

39)

Observe that

$$\frac{d}{dx}\left(\frac{x}{\ln x}\right) = \frac{\ln x - 1}{(\ln x)^2}.$$

Thus

$$\int_3^\infty \frac{\ln x - 1}{(\ln x)^2}\,dx = \lim_{a\to\infty}\left[\frac{x}{\ln x}\right]_3^a = \lim_{a\to\infty}\frac{a}{\ln a} - \frac{3}{\ln 3}.$$

By L'Hôpital's Rule, $\lim_{a\to\infty} a/\ln a = \lim_{a\to\infty} 1/(1/a) = +\infty$, so the integral diverges.